Computing the k-universal transversal property

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Lisbon, 11 May 2026

Introduction

- Over many years, João Araújo, Wolfram Bentz, Peter Cameron, and their colleagues have been investigating properties of semigroups of the form

$$\langle G, t \rangle,$$

where G is a permutation group on a finite set Ω, and

$$t : \Omega \to \Omega$$

is a non-permutation, in terms of properties of the group G.

- These investigations have provided me with interesting algorithmic and computational challenges, which I have greatly enjoyed, so many thanks go out from me to João, Wolfram, and Peter!

The k-universal transversal property

Today, I want to talk about the k-universal transversal property of a permutation group G, and the computational determination of this property.

Definition

Let G be a permutation group on a finite set Ω of size n, and let k be an integer with $2 \leq k \leq n$. We say that G satisfies the k-**universal transversal** property (or k-**ut** for short) if for every k-subset A of Ω and every k-partition P of Ω there is a $g \in G$ such that A^g is a transversal of P.

k-ut

- In other words, G satisfies k-ut if and only if for every partition $P = \{P_1, P_2, \ldots, P_k\}$ of Ω into exactly k parts, every G-orbit of subsets of size k of Ω contains at least one element $\{a_1, a_2, \ldots, a_k\}$ such that $a_i \in P_i$ for $i = 1, 2, \ldots, k$.

- It is easy to see that if G is a k-homogeneous permutation group on Ω, that is, transitive on the k-subsets of Ω, then G satisfies k-ut.

- It is also straightforward to show that the 2-ut property is equivalent to primitivity of the permutation group G on Ω (when $|\Omega| > 2$).

The k-ut property's application to semigroups

The importance of the k-ut property to semigroup theory is the following:

Theorem (J. Araújo and P.J. Cameron)

Let G be a permutation group on a finite set Ω of size $n \geq 5$ and let k be an integer with $2 \leq k \leq \lfloor (n+1)/2 \rfloor$. Then G satisfies k-ut if and only if for every transformation $t : \Omega \to \Omega$ of rank at most k (that is, the image of t has size at most k), the semigroup $T := \langle G, t \rangle$ is regular (that is, for every $a \in T$ there is a $b \in T$ such that $a = aba$).

The algorithm and its implementation

- An approach to test if a given permutation group satisfies the k-ut property was suggested to me by João Araújo.
- I have modified and developed this approach, and implemented it in the open-source algebraic system GAP, also using its package GRAPE (for classifying orbits on sets, and computing efficiently with compact graph data structures), together with an optional C program.
- The fully documented programs form part of the open-source repository
  https://github.com/lhsoicher/transversalproperties/
  which contains functions to compute the k-ut, "k-et", and "strong k-ut" properties of a permutation group.

Remark on the implementation of a G-orbit on k-sets

A G-orbit $\mathcal{O}$ of k-subsets of Ω is implemented as a directed graph with vertex set $\Omega \cup \binom{\Omega}{k-1}$ and with (x, y) a directed edge if and only if $x \in \Omega$ and y is a $(k-1)$-subset of Ω such that $\{x\} \cup y$ is in the orbit $\mathcal{O}$.

Such a graph is computed and stored efficiently using the compact GRAPE package graph data structure, and allows the rapid determination of the k-subsets in $\mathcal{O}$ containing a given point in Ω.

The function `transversalproperty`

- The heart of my algorithm to determine the k-ut property is a recursive function called `transversalproperty`, which is implemented in both GAP and C, such that either implementation can be called from within GAP.
- I will outline a simplified version of this function, where as usual, G is a permutation group on a finite set Ω of size n, and k is an integer with $2 \leq k \leq n$.

`transversalproperty`: input and return value

Given a G-orbit $\mathcal{O}$ of k-subsets of Ω (represented compactly as described above) and an ordered partition $P = (P_1, \ldots, P_k)$ of Ω such that $|P_k| \geq n/k$, the function `transversalproperty` attempts to build an ordered partition $Q = (Q_1, \ldots, Q_k)$ of Ω such that:

- $P_i \subseteq Q_i$ for $i = 1, \ldots, k-1$,
- $|Q_k| \geq n/k$, and
- no k-subset in the G-orbit $\mathcal{O}$ is a transversal of Q.

The function either returns such a partition Q, or shows that none exists and returns the value `true`.

`transversalproperty`: the main steps

1. For each $A \in \mathcal{O}$ that is a transversal of P, the element $a \in A$ that is in P_k cannot be in Q_k, and we build a set R consisting of these elements a (our data structures and further function parameters allow this R to be determined efficiently).

2. If R is the empty set, there is no transversal of P in $\mathcal{O}$, in which case we are done, and the function returns P.

3. If $|P_1| + \cdots + |P_{k-1}| + |R| > n(k-1)/k$ then we also are done (as there is no Q with the required properties), and the value `true` is returned.

4. Otherwise, an element r is removed from R and the function is called recursively with $\mathcal{O}$ and $P_i(r)$, for $i = 1, \ldots, k-1$, where $P_i(r)$ is formed from P by removing r from P_k and adding it to P_i. If any such recursive function call returns a partition Q then this Q is returned by the present function call. Otherwise the present function call returns `true`.

The main algorithm: input and return value

- The main algorithm to decide the k-ut property takes as input a permutation group G on a finite set Ω of size n and an integer k with $2 \leq k \leq n$.
- The algorithm returns `true` if G satisfies k-ut and `false` if not. (In the case where the answer is `false`, the algorithm can provide a k-subset A and k-partition Q of Ω such that no G-image of A is a transversal of Q.)

The main algorithm's main steps

1. The algorithm begins by determining whether G is k-homogeneous, and if so, returns `true`. Otherwise, (lex-least) representatives $A_1, \ldots, A_u$ for the G-orbits on the k-subsets of Ω are determined (alternatively, N-orbits for a given N where $G \leq N \leq N_{\mathrm{Sym}(\Omega)}(G)$).

2. The algorithm then calculates (lex-least) representatives $B_1, \ldots, B_v$ for the G-orbits on $(k-1)$-subsets of Ω. Now if for some i and j, A_i does not contain a G-image of B_j, the algorithm returns `false` (since for such an A_i, B_j, if $B_j = \{b_1, \ldots, b_{k-1}\}$ then the G-orbit of A_i does not contain a transversal of the partition $\{\{b_1\}, \{b_2\}, \ldots, \{b_{k-1}\}, \Omega \setminus B_j\}$).

The main algorithm's main steps (continued)

3. For each k-subset representative $A \in \{A_1, \ldots, A_u\}$ the main algorithm performs the the next step to determine whether the G-orbit of A contains a transversal for every k-partition P of Ω. This is done in an order of non-increasing size of the G-stabilizer of A (since shorter orbits of k-subsets are thought more likely to fail the test, in which case the algorithm can return `false` and stop).

4. To test the G-orbit $\mathcal{O}$ of a k-subset A, the function `tranversalproperty` is called with $\mathcal{O}$ (suitably represented) and the ordered k-partition $(\{b_1\}, \{b_2\}, \ldots, \{b_{k-1}\}, \Omega \setminus B)$, for each $B = \{b_1, \ldots, b_{k-1}\} \in \{B_1, \ldots, B_v\}$.

5. If `transversalproperty` returns `true` for every A and B, the main algorithm returns `true`; otherwise it returns `false`.

4-ut

- The permutation groups of degree n satisfying k-ut have been completely classified for all k such that $5 \leq k \leq \lfloor (n+1)/2 \rfloor$, by Araújo and Cameron.

- The permutation groups satisfying 4-ut are also known, except possibly for the groups P acting on the $q+1$ points of the projective line $\mathrm{PG}(1,q)$ and satisfying:
  - $\mathrm{PSL}(2,q) \leq P \leq \mathrm{P\Gamma L}(2,q)$,
  - q is a prime with $q \equiv 11 \pmod{12}$ or $q = 2^m$ with m prime,
  - for all $c \in \mathrm{GF}(q) \setminus \{0,1\}$ the elements $-1, c, c-1$ generate the multiplicative group of the field $\mathrm{GF}(q)$.

- It had previously been checked that these P satisfy 4-ut for $q \leq 50$.

Some 4-ut computations

- I further checked that these P satisfy 4-ut for $q \leq 250$, applying my programs to all $\mathrm{PSL}(2, q)$ with $q \leq 250$ and either q a prime with $q \equiv 11 \pmod{12}$ or $q = 2^m$. This required about 24 hours of CPU time.

- Satisfyingly, the groups tested failed 4-ut only for

$$q \in \{16, 64, 131, 191, 239\},$$

that is, for either $q = 2^m$ with m composite or for some $c \in \mathrm{GF}(q) \setminus \{0, 1\}$ the elements $-1, c, c - 1$ do not generate the multiplicative group of $\mathrm{GF}(q)$.

Some 3-ut computations

- I have similarly verified that when $q \leq 200$, and either q is a prime with $q \equiv 11 \pmod{12}$ or $q = 2^m$ with m prime, that the groups A of index at most 2 in $\mathrm{AGL}(1, q)$ (acting on the q points of the affine line $\mathrm{AG}(1, q)$) satisfy 3-ut if and only if for all $c \in \mathrm{GF}(q) \setminus \{0, 1\}$ the elements $-1, c, c - 1$ generate the multiplicative group of $\mathrm{GF}(q)$.

- The situation for $q > 200$ still needs to be dealt with.

Does $\mathrm{Sz}(2^{2m+1})$ satisfy 3-ut?

- It is unknown whether the 2-transitive simple groups $\mathrm{Sz}(2^{2m+1})$ of Suzuki, with permutation degree $(2^{2m+1})^2 + 1$, satisfy 3-ut for all integers $m \geq 1$.
- Pablo Spiga has determined by hand that $\mathrm{Sz}(8)$ does satisfy 3-ut.
- I have confirmed this computationally and have computed that $\mathrm{Sz}(32)$ (of degree 1025) also satisfies 3-ut. The latter computation required only about 15 seconds of CPU time.
- Beyond that, nobody knows!

The k-existential transversal property

It was easy to adapt my k-ut programs to handle also the significantly weaker k-et property.

Definition

Let G be a permutation group on a finite set Ω of size n, and let k be an integer with $2 \leq k \leq n$. We say that G satisfies the k-**existential transversal** property (or k-**et** for short) if there is a k-subset A of Ω such that, for every k-partition P of Ω there is a $g \in G$ for which A^g is a transversal of P.

Some k-et computations

- The most important computation of k-et that I have done shows that $\mathrm{P\Gamma L}(2, 128)$ on 129 points satisfies 5-et, completing the classification of the groups satisfying 5-et by Araújo, Bentz, and Cameron.

- I have also shown that the following groups satisfy 4-et:
  - $\mathrm{Sz}(8)$ on 65 points,
  - $\mathrm{PGU}(3, 5)$ on 126 points,
  - HS on 176 points,
  - $\mathrm{PSU}(3, 8).3^2$ on 513 points, and
  - $\mathrm{PSL}(2, q)$ on $q + 1$ points for q up to 400.

- However, $\mathrm{PSU}(3, 8).6$ on 513 points fails 4-et.

Further challenges

There are still further difficult challenges, such as deciding 4-et for:

- a certain $\mathrm{PSU}(3,8).3$ on 513 points,
- $\mathrm{P\Gamma U}(3,9)$ on 730 points,
- $\mathrm{Sz}(32).5$ on 1025 points,
- $\mathrm{P\Gamma U}(3,16)$ on 4097 points, and
- $\mathrm{PSL}(2,q)$ on $q+1$ points for $q > 400$.

More ideas are needed!